

**Министр науки и высшего образования Российской
Федерации**

**Федеральное государственное автономное
образовательное учреждение высшего образования
«Национальный исследовательский университет
ИТМО»**



**Факультет информационных технологий и
программирования**

Лабораторная работа № 02

Лабораторная работа по Git

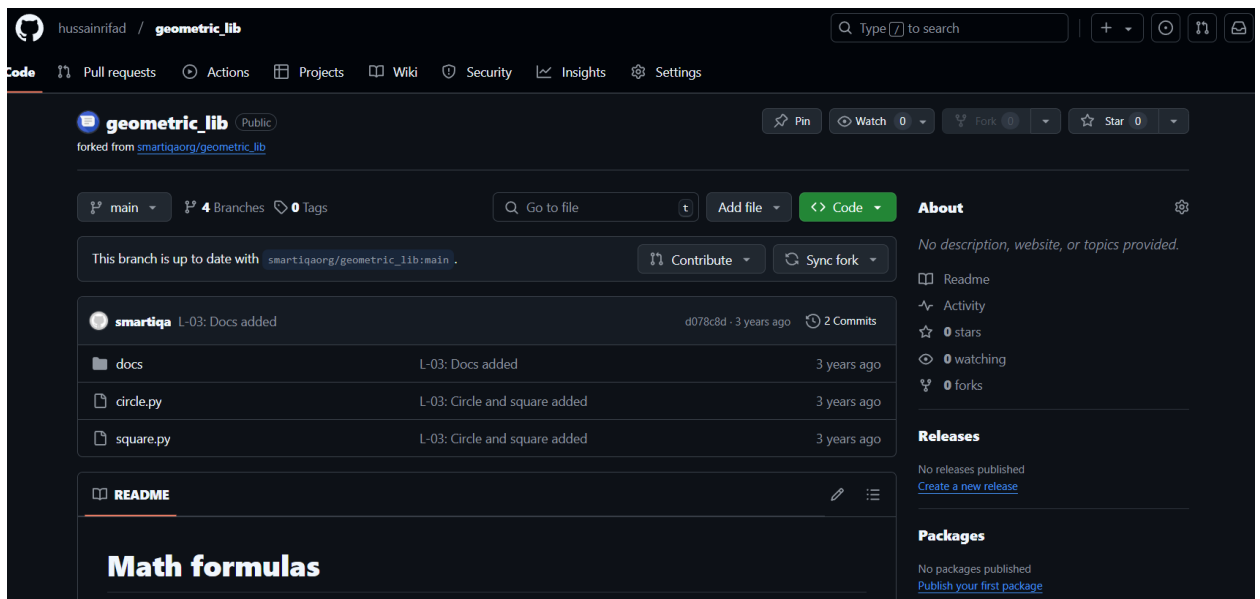
**Выполнил студент группы № М3113
МД Абу Хуссаин Рифад**

**Проверил:
Жуйков Артём Сергеевич**

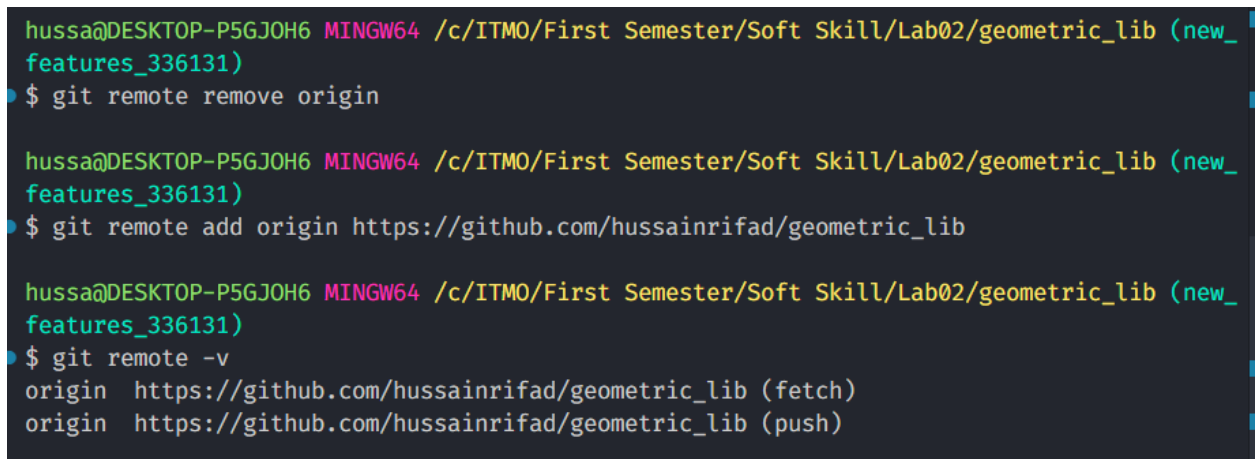
Санкт-Петербург
2024

Part-1

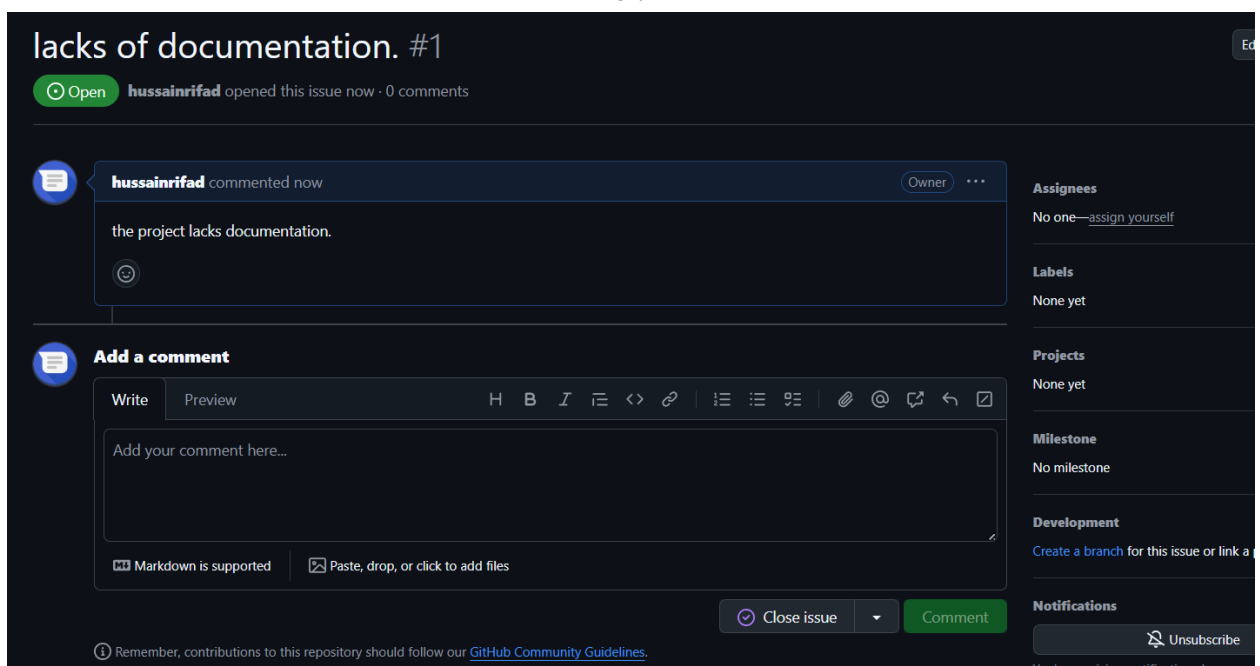
1.



2.



3.



Part-2

1.

```
hussa@DESKTOP-P5GJOH6 MINGW64 /c/ITMO/First Semester/Soft Skill/Lab02/geometric_lib (new_
features_336131)
$ git branch
  documentation
  main
* new_features_336131
```

3.

```
# Geometric Library

This Python code defines a library of functions for calculating the area
and perimeter of various geometric shapes. It includes calculations for
circles, rectangles, squares and triangles.

## Functions

### Circle Calculations

#### `area(r)`
Calculates the area of a circle.

* **Parameters:**
  * `r`: The radius of the circle (a numerical value).
* **Returns:**
  * The area of the circle (a numerical value).
* **Example:**
  ```python
 import math
 area_of_circle = area(5)
 print(f"The area of a circle with radius 5 is: {area_of_circle}") #
 Output: The area of a circle with radius 5 is: 78.53981633974483
  ```

#### `perimeter(r)`
Calculates the perimeter (circumference) of a circle.

* **Parameters:**
  * `r`: The radius of the circle (a numerical value).
* **Returns:**
  * The perimeter of the circle (a numerical value).
* **Example:**
  ```python
 perimeter_of_circle = perimeter(5)
 print(f"The perimeter of a circle with radius 5 is:
 {perimeter_of_circle}") # Output: The perimeter of a circle with
 radius 5 is: 31.41592653589793
```

```

Square Calculations

`area(a)`
Calculates the area of a square.

* **Parameters:**
 * `a`: The length of one side of the square (a numerical value).
* **Returns:**
 * The area of the square (a numerical value).
* **Example:**
  ```python
  area_of_square = area(5)
  print(f"The area of a square with sides of 5 is: {area_of_square}") #
  Output: The area of a square with sides of 5 is: 25
  ```

`perimeter(a)`
Calculates the perimeter of a square.

* **Parameters:**
 * `a`: The length of one side of the square (a numerical value).
* **Returns:**
 * The perimeter of the square (a numerical value).
* **Example:**
  ```python
  perimeter_of_square = perimeter(5)
  print(f"The perimeter of a square with sides of 5 is:
  {perimeter_of_square}") # Output: The perimeter of a square with sides
  of 5 is: 20
  ```

Triangle Calculations

```

## Part-3

1.

```

$ git add .

hussa@DESKTOP-P5GJOH6 MINGW64 /c/ITMO/First Semester/Soft Skill/Lab02/geometric_lib (new_
features_336131)
$ git commit -m"documantation updated"
[new_features_336131 a9c77b4] documantation updated
1 file changed, 127 insertions(+), 10 deletions(-)

hussa@DESKTOP-P5GJOH6 MINGW64 /c/ITMO/First Semester/Soft Skill/Lab02/geometric_lib (new_
features_336131)
$ git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 1.29 KiB | 660.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/hussainrifad/geometric_lib
1d86eb5..a9c77b4 new_features_336131 -> new_features_336131

```

2 & 3.

## New features 336131 #1306

Open **hussainrifad** wants to merge 2 commits into `smartigaorg:main` from `hussainrifad:new_features_336131`

Conversation 0

Commits 2

Checks 0

Files changed 3

+138 -10

**hussainrifad** commented now

This PR adds some documentation to the project.

Addresses issue [hussainrifad#1](#)

**@Mindtable** please review when you have a chance.

hussainrifad added 2 commits last month

new file has been added1d86eb5

documentation updateda9c77b4

✓

This branch has no conflicts with the base branch

Only those with [write access](#) to this repository can merge pull requests.

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development